

MIRPUR UNIVERSITY OF SCIENCE AND TECHNOLOGY



Department of Computer Science and Information Technology FINAL YEAR PROJECT PROPOSAL & PLAN

TutorPk

Project Team

Name of Student	Reg. Numbers	Program	Contact Number	Email Address
Noman Sabir	FA22-BCS-006	BSCS	0326-0857571	nomansabir8635@gmail.com
Muhammad Husnain	FA22-BCS-012	BSCS	0331-5454947	za8644275@gmail.com
Suleman Ahmad	FA22-BCS-029	BSCS	0346-5302787	sulemanali67433@gmail.com

Acceptance by Supervisor

I have read the proposal and agree to supervise above project team for partial requirement of the degree program _____. The proposed project is academically, logistically and financially feasible and all the required literature, equipment and laboratories facilitates are or will be available.

Project Supervisor (Designation): _____

Date: _____

TutorPk

Change Record

[illegible]

Project Proposal

Project Title: TutorPk

Introduction

Online education has become increasingly important in Pakistan due to the growth of digital learning and limited access to quality education in many areas. Students from schools, colleges, and universities often depend on private tutors for academic support in various subjects. However, finding reliable and qualified tutors is a challenge, especially for students living in remote or underserved regions. On the other hand, many capable tutors struggle to reach students due to the lack of a proper online platform. TutorPk is being developed for students and tutors across Pakistan to provide a structured and accessible online tutoring system. The platform aims to facilitate effective communication, organized class management, and academic coordination in a single centralized environment tailored to local educational needs.

Existing System / Description of the Current Situation

Online tutoring in Pakistan is mostly informal, using WhatsApp, Facebook, and personal contacts, with no proper verification or organization. Students cannot reliably assess tutor quality, tutors face scheduling and coordination issues, and payments are handled informally, making professional and effective learning difficult.

Sabaq Foundation Limitations:

- No support for live tutoring or real-time interaction between students and teachers.
- Lacks personalized tutor recommendations.
- No scheduling system for lessons or sessions.
- Does not provide individual student progress tracking.
- Limits interactive and adaptive learning experiences.

Knowledge Platform Limitations:

- No live classes or real-time teaching options.
- Lacks scheduling modules for sessions.
- No direct communication between students and tutors.
- No intelligent recommendations for tutors or learning materials.
- Absence of gamification features to enhance student motivation and engagement.

Problem Statement

The absence of a centralized and structured online tutoring platform in Pakistan has resulted in unorganized learning practices and reduced educational quality. Students face difficulties in finding verified tutors, managing class schedules, and maintaining consistent communication. Tutors struggle with handling multiple students, organizing sessions, and

maintaining professional records. The lack of academic tracking, proper scheduling, and reliable communication creates inefficiencies for both parties. Informal systems also fail to provide motivation and engagement for students. These challenges highlight the need for a reliable, organized, and accessible online tutoring solution that supports both learners and tutors in a professional manner.

Proposed Solution

TutorPk proposes a smart, recommendation-driven tutoring platform developed to support students and tutors across Pakistan's academic landscape. The system will connect students with verified tutors through structured profiles and subject-based categorization. It will provide dedicated dashboards for students, tutors, and administrators to manage communication, academic records, and class schedules. A scheduling module will allow tutors to create class sessions within platform (Jitsi API) or share links of zoom/ Team with students. The platform will support secure subscription-based payments and include a chat system for smooth interaction and notifications. Basic recommendation logic will help students find suitable tutors, while gamification features will encourage student engagement. Administrators will ensure platform quality through verification and monitoring tutor and can block them in case of Reporting from students.

Scope of the Project

User Management Module

This module will handle registration and login for students, tutors, and administrators. Users will be able to manage their profiles, including personal information and academic details. Tutors will submit profiles for verification, which will be approved by the administrator. This module ensures secure and role-based access to the system.

Tutor Search and Recommendation Module

Students will be able to search tutors based on subjects and basic preferences. The system will provide simple, rule-based recommendations using factors such as subject selection and tutor popularity. This module helps students find suitable tutors efficiently without complex AI models.

Class Scheduling and Communication Module

Tutors will create class schedules and share session details with enrolled students. A built-in chat system will allow communication between students and tutors. Students will receive notifications regarding class updates and schedules through the system.

Payment and Subscription Module

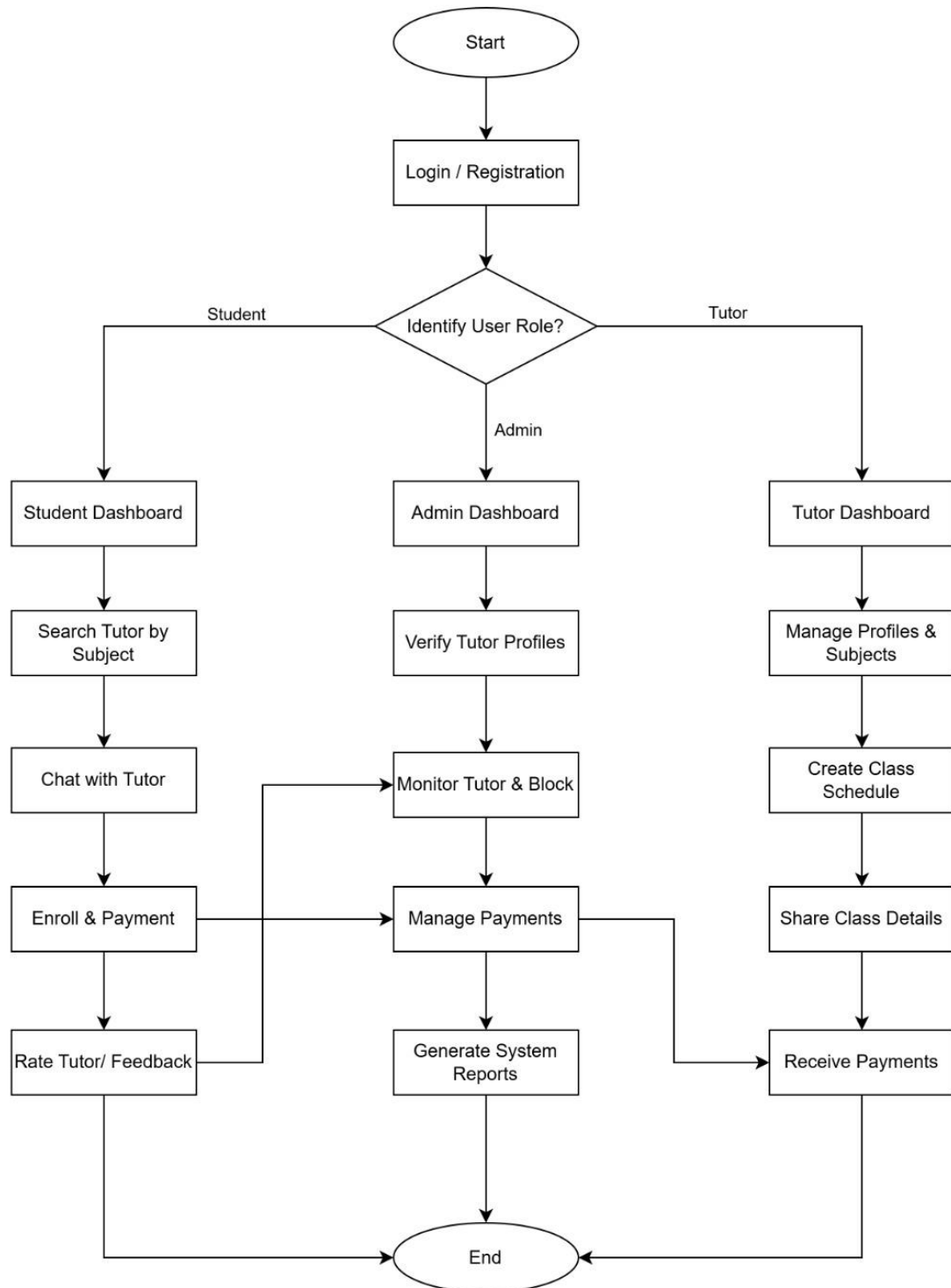
This module will manage subscription-based tutoring payments. Students will be able to subscribe to tutoring services, and payment records will be stored for reference. The system will ensure a basic and secure payment flow suitable for the project scope.

Administrators will verify tutors, and oversee system activities. This module will help maintain platform quality, manage subjects, and monitor overall usage. Basic reports and records will support administrative decision-making and block tutor based on bad reviews.

Supervisor's Signature: _____
Co-Supervisor's Signature (If applicable): _____

[illegible]

Flow Chart



Roles & Responsibility Matrix:

The purpose of roles & responsibility matrix is to identify who will do what.

WBS #	WBS Deliverable	Activity #	Activity to Complete the Deliverable	Duration (Days)	Responsible Team Member(s) & Role(s)
1.1	Work Breakdown Structure	1.1.1	Identify project phases and major tasks	2	All Members – Planning
		1.1.2	Define deliverables for each phase	2	All Members – Planning
1.2	Roles & Responsibility Matrix	1.2.1	Assign frontend, backend, database roles	1	Team Lead
		1.2.2	Document responsibility matrix	2	Noman Sabir – Documentation
1.3	Change Control System	1.3.1	Define change request procedure	1	Muhammad Husnain – Analysis
		1.3.2	Approval mechanism with supervisor	1	All Members
2.1	Documentation Introduction	2.1.1	Write documentation overview	1	Noman Sabir – Documentation
2.2	Literature / Market Survey	2.2.1	Study existing tutoring platforms	3	Muhammad Husnain – Research
		2.2.2	Analyze informal tutoring systems in Pakistan	2	Suleman Ahmed – Research
2.3	Requirements Analysis	2.3.1	Identify functional requirements	3	Muhammad Husnain – Analysis
		2.3.2	Identify nonfunctional requirements	2	Suleman Ahmed – Analysis

2.4	System Design	2.4.1	Design system architecture	3	Suleman Ahmed – Backend
		2.4.2	Design database schema	5	Muhammad Husnain – Database
2.5	Implementation	2.5.1	Frontend development	15	Noman Sabir – Frontend

		2.5.2	Backend development	30	Suleman Ahmed – Backend
		2.5.3	Database implementation	10	Muhammad Husnain – Database
2.6	System Reports	2.6.1	Generate user and class reports	2	Muhammad Husnain – Reports
2.7	Testing & Performance Evaluation	2.7.1	Module testing	5	All Members – Testing
		2.7.2	Performance evaluation	3	Suleman Ahmed – Testing
2.8	Conclusion & Outlook	2.8.1	Write conclusion and future work	3	Noman Sabir – Documentation
2.9	End User Documentation	2.9.1	Prepare student & tutor user guide	2	Noman Sabir – Documentation
2.10	Application Administration Documentation	2.10.1	Admin operation documentation	3	Muhammad Husnain – Documentation
2.11	System Administrator Documentation	2.11.1	System setup & maintenance guide	3	Suleman Ahmed – Documentation
3.1	Development Environment	3.1.1	Setup IDE and local server (XAMPP)	2	Suleman Ahmed – Backend
		3.1.2	Configure MySQL database	5	Muhammad Husnain – Database
3.2	Interface / Presentation Layer	3.2.1	Student interface development	7	Noman Sabir – Frontend
		3.2.2	Tutor interface development	7	Noman Sabir – Frontend

		3.2.3	Admin interface development	5	Noman Sabir – Frontend
3.3	Business Logic Layer	3.3.1	Authentication & role management	3	Suleman Ahmed – Backend
		3.3.2	Tutor recommendation logic	3	Muhammad Husnain – Backend
		3.3.3	Scheduling & subscription logic	6	Suleman Ahmed – Backend
3.4	Data Management Layer	3.4.1	User & profile tables	2	Muhammad Husnain – Database
		3.4.2	Class & subscription records	2	Muhammad Husnain – Database
3.5	Physical Layer	3.5.1	Local deployment	2	Suleman Ahmed – Deployment
3.6	System Reporting	3.6.1	Registration reports	1	Muhammad Husnain
		3.6.2	Tutor activity reports	3	Muhammad Husnain
		3.6.3	Class & payment reports	4	Muhammad Husnain

Approval

Project Supervisor

Comments _____

Name: _____ Date: _____

Signature: _____

Project Co-Supervisor (If Applicable)

Comments _____

Name: _____ Date: _____

Signature: _____

Project Coordinator

Comments _____

Name: _____ Date: _____

Signature: _____